

ANNEX

GUIDELINES FOR THE PREPARATION OF THE REPORTS

FROM MEMBER STATES ON THE IMPLEMENTATION OF THE

2021 RECOMMENDATION ON OPEN SCIENCE

I. INTRODUCTION

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as “the 2021 Recommendation”) that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i.e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO’s Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

II. WHAT ARE THE AIMS OF THIS CONSULTATION?

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

III. HOW TO FILL IN THE QUESTIONNAIRE?

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organise the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at <https://www.unesco.org/en/open-science/toolkit> and <https://unesdoc.unesco.org/ark:/48223/pf0000387324>.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

- (i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted [online](#).
- (ii) Email: the questionnaire can be completed electronically and sent to: openscience@unesco.org

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

IV. QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country*: Austria
- Organization(s) or entity(ies) responsible for the preparation of the report*:

Name: Austrian Commission for UNESCO

Website: www.unesco.at

Email address*:

Phone number*:

Please describe the role/mandate of your organisation: The Austrian Commission for UNESCO (ÖUK) is the national coordination and liaison office for UNESCO.

- Officially designated contact person(s) who completed this survey:

Full name*: Ms. Christine Maaß

Position*: Head of Department of Science and Research

Email address*:

- Full Name(s) of designated official(s) certifying the report: Mr. Nikolaus Possanner (Federal Ministry of Women, Science and Research)
- Brief description of the consultation process established for the preparation of the report:

The preparation of the Austrian report was coordinated by the Austrian Commission for UNESCO, in collaboration with the Federal Ministry of Education, Science, and Research. Representatives of organisations within the broader Austrian open science ecosystem, such as higher education institutions, science funding agencies and civil society organisations, were also consulted for the preparation of this report. Statistical and other data included in the report were provided by the Federal Statistics Office (Statistics Austria). In addition, the Austrian Commission for UNESCO conducted desk research to identify further relevant information and data. Based on a consolidated draft, the final report was prepared by the Federal Ministry of Education, Science, and Research. As of 1 April 2025, responsibility for science and research agendas was transferred to the newly established Federal Ministry of Women, Science and Research, which also certified this report.

Other organisation (s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organisation*: Federal Ministry of Education, Science, and Research; Agency for Education and Internationalisation (Sparkling Science 2.0 research funding programme); Austrian Science Fund; Federal Statistical Office; Association of Austrian Adult Education Centres; Austrian Academy of Sciences (FAEKT, science communication); BOKU Vienna (Institute of Zoology); Citizen Science Network Austria; Diskurs. Das Wissenschaftsnetz; Open Science Austria; TU Wien (Center for Research Data Management); University of Graz (Center for Society, Knowledge and Communication); University of Veterinary Medicine Vienna (University Library); University of Vienna

Website*:

<https://www.bmfwf.gv.at/en>, <https://oead.at/en/study-research-teaching/citizen-science/sparkling-science>, <https://www.fwf.ac.at/en/about-us/what-we-do/open-science>, <https://www.statistik.at/en/>, <https://www.vhs.or.at/english>, <https://www.oewar.ac.at/en/funding/funding-programmes>, <https://boku.ac.at/en/oekb/zool>, <https://www.citizen-science.at/en/network>, <https://www.diskurs-wissenschaftsnetz.at/>, <https://www.osa-openscienceaustria.at/en/>, <https://www.tuwien.at/en/research/rti-support/research-data/9>, <https://sieben.uni-graz.at/de/>, <https://www.vetmeduni.ac.at/en/bibliothek>, <https://www.univie.ac.at/en/>

Email address*:

Sector: ☐ **Public** ☐ Private ☐ **Civil Society** ☐ Other

B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organisations in your country?

YES / NO

If yes, please indicate which ministries and/or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

UNESCO's 2021 Recommendation was shared with the Federal Ministry responsible for Education, Science and Research. A broad range of other stakeholders – e.g. at research and higher education institutions – are aware of the Recommendation.

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES / NO

If yes, which language(s)?

The 2021 Recommendation was translated into German in cooperation between the UNESCO Commissions of Germany, Austria, Switzerland and Luxembourg, and was published by the German Commission for UNESCO in 2024: https://www.unesco.at/fileadmin/user_upload/UNESCO-Empfehlung_zu_Open_Science_Deutsche_Uebersetzung_01.pdf. With regard to the languages of autonomous population groups recognised as official languages in some Austrian regions, the Recommendation is also available in Hungarian: https://unesco.hu/data/Recommendation_on_Open_Science_hun_final.pdf.

1.3 Have there been awareness raising activities on open science, including all its key elementsⁱ, associated benefits and challenges, organised or foreseen to be organised by the end of 2025 in your country by national authorities or entities? (ref.: [\(i\) 16.f, 16.g, 16.h](#))

YES / NO

If yes, please provide details on the activities organised or foreseen and links as relevant.

In Austria, various activities related to the promotion of open science have been carried out between 2021 and 2024, with additional initiatives planned for 2025. Examples include: (1) In October 2023, an international conference themed "5 Years of EOSC Developments", organised by the Federal Ministry responsible for Education, Science and Research at the University of Vienna, presented perspectives, opportunities and challenges for data in science and research in Europe. A recording of the conference is available at: <https://eosc-lustrum.eu/home/>; (2) The Austrian Citizen Science Conference has been the main annual event for the citizen science community in Austria since 2015. Each year, the conference is hosted by a different institution in close collaboration with "Österreich forscht". The focus of the first two days of the conference is on a scientific programme, while the third day invites citizen scientists and the public to discover new citizen science projects: <https://www.citizen-science.at/en/conference>; (3) The University of Vienna will organise an Open Science Festival in September 2025 to explore the future of open science through keynotes and panel

discussions. The event will also provide networking opportunities for researchers, students, and research support staff: <https://osfestival2025.univie.ac.at/en/>

- 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open scienceⁱⁱ, in publicly funded research? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))**

YES / NO

If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

Austria has specific funding mechanisms at the national level to support open science practices. These mechanisms ensure compliance with open science principles such as open access, data sharing, and transparency in research processes. For more details, see sections 2.6 and 2.7 of this report.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

- 2.1 Does your country have a national policyⁱⁱⁱ, strategy or plan of action on science, technology and innovation (STI)?**

YES / NO

If yes, please attach/upload a machine-readable PDF file of the policy and provide the following information: Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.

Austrian RTI Strategy 2030 – RTI Strategy 2030 Strategy for Research, Technology and Innovation of the Austrian Federal Government / The top level of the RTI Task Force is composed of representatives from the Federal Chancellery, the Federal Ministry of Finance, the Federal Ministry of Education, Science and Research, the Federal Ministry for Digital and Economic Affairs, and the Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology. / There was a broad stakeholder consultation online, including universities, research institutes, private enterprises, social partners, research promotion agencies and scientific funds, schools, federal states etc. / 2020 / - / The RTI Strategy sets out the strategic direction for ten years through a set of overarching objectives: (1) become an international leader in innovation and strengthen Austria as an RTI location, (2) focus on effectiveness and excellence, and (3) focus on knowledge, talent and skills. The strategy refers to open science as one of several cross-cutting issues on which it is based. / https://era.gv.at/public/documents/4489/RTI_Strategy_2030-1-1.pdf, <https://era.gv.at/policies/austrian-rti-strategy-2030/>

- 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)^{iv} at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: [\(i\) 17.a, b, c, d, e, h](#))**

YES / PARTLY / NO

If yes or partly, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.

(1) Open Science Policy Austria (“Open Science Austria – Österreichische Policy zu Open Science und der European Open Science Cloud”) / The policy was collaboratively developed by the Federal Ministry of Education, Science and Research, the Federal Ministry for Digital and Economic Affairs, and the Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology / - / 2022 / - / Publication of scientific and research data based on the FAIR Principles, Participation in European and International Open Science Processes – the European Open Science Cloud (EOSC), Open Access – Access to Publicly Funded Publications, Open Educational Resources (OER) / Open access to scientific knowledge, open science infrastructure, open engagement of societal actors / <https://www.bmfwf.gv.at/wissenschaft/leitthemen/digitalisierung/open-science/open-science-policy-austria.html>. More information on the underlying legal framework can be found in the country report prepared by a working group within the EOSC Support Office Austria: <https://eosc-austria.at/working-groups/austria-country-report/> (Q3/2024).

(2) Austrian Data Strategy (“Datenstrategie für Österreich”) / Federal Chancellery of Austria / - / 2024 / - / The strategy pursues three goals: (1) Develop sustainable data infrastructures and technical solutions for efficient data exchange, (2) Unlock the potential for responsible data sharing, (3) Establish an innovative data culture and increase data literacy / Open access to scientific knowledge, open science infrastructure, open engagement of societal actors / https://www.digitalaustria.gv.at/dam/jcr:4cca42d8-3112-4a9e-bb52-822c4adecea0/Datenstrategie_barrierefrei_final-02102024.pdf (in German).

(3) Austria’s E-Government Strategy 2023 (“E-Government Strategie Österreich 2023”) / Federal Ministry of Finance / The strategy was jointly produced with over 80 representatives of the federal government, provinces, municipalities and local communities / 2023 / - / The e-government strategy establishes a uniform, networked and coordinated approach to e-government through intensive cooperation between the federal government, federal states, cities and municipalities. As target groups for administrative action, the focus is primarily on citizens and companies. / Open access to scientific knowledge, open science infrastructure / <https://www.digitalaustria.gv.at/dam/jcr:e93f9913-c1d1-4968-b6c6-ea9aed5c3498/E-GovernmentStrategieOesterreich-EN-final17072023.pdf>.

(4) Intellectual Property Strategy for Austria (“IP-Strategie für Österreich – Strategie der österreichischen Bundesregierung für geistiges Eigentum”) / Federal Ministry of Science, Research and Economy, Federal Ministry for Transport, Innovation and Technology / The Federal Ministry of Education, the Federal Ministry of Finance, the Federal Ministry of Justice, and the Federal Chancellery supported the process. Representatives from universities, research institutions and companies were also involved. An international reflection group accompanied the creation of the strategy. / 2017 / - / Legal framework, institutional framework, skills, targeted support for innovators and creative professionals in the effective use of intellectual property rights, interfaces with other strategies of the Federal Government / Open access to scientific knowledge, open science infrastructure / https://www.patentamt.at/fileadmin/root_oepa/Dateien/Presse/20170214_iP_Barrierefrei_.pdf (in German).

(5) Open Innovation Strategy for Austria (“Open Innovation Strategie für Österreich”) / Federal Ministry of Science, Research and Economy, Federal Ministry for Transport, Innovation and Technology / The policy was drawn up in a broad participatory process involving stakeholders from research, science and business, as well as interested members of the public / 2016 / - / The strategy provides an analysis of the national and international status quo on open innovation while providing a vision for 2025 and concrete measures for implementation. Measure 12 of the Open Innovation Strategy e.g. is tackling how to embed principles of open data and open access in research / open access to scientific knowledge / https://openinnovation.gv.at/wp-content/uploads/2015/08/OI_Barrierefrei_Englisch.pdf

2.3 In your country, are there specific policy instruments^v that aim to promote open science? your country, is there a policy, or a set of policies and/or legal framework(s)^{vi} at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science?

YES/ UNDER DEVELOPMENT/NO

If yes or under development, please provide details and links as relevant, including for each instrument: Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.

(1) RTI Pact 2024–2026 (“FTI-Pakt 2024–2026”) / The RTI Pact operationalises the targets and areas of activity of the RTI Strategy 2030 and defines strategic research and innovation priorities. The Federal Ministries will subsequently implement these priorities through performance and funding agreements with central institutions, in line with their respective statutory mandates and through further measures as necessary (2024–2026) / Federal Government of the Republic of Austria / - / e.g. open engagement of societal actors / The document is available for download at: <https://www.bundeskanzleramt.gv.at/en/topics/rti-strategy.html>.

(2) RTI Pact 2021–2023 (“FTI-Pakt 2021–2023”) / The RTI Pact operationalised the targets and fields of activity of the RTI Strategy 2030 and defined strategic research and innovation priorities, to be implemented by the Federal Ministries through performance and funding agreements with the central institutions in line with their respective statutory mandates (2021–2024). / Federal Government of the Republic of Austria / - / Open science infrastructure, in particular the European Open Science Cloud (EOSC) Partnership, open engagement of societal actors / The document is available for download at: <https://www.bundeskanzleramt.gv.at/en/topics/rti-strategy.html>.

(3) Research Infrastructure Action Plan 2030 (“Forschungsinfrastruktur-Aktionsplan 2030”) / The action plan provides a development perspective until 2030 in line with Austria's RTI Strategy 2030. The guiding principle of the action plan is the coordinated procurement and cooperative use of research infrastructures. / An interministerial RTI task force is responsible for coordinating RTI policy at the federal level and managing and supporting the implementation of the RTI strategy. The task force is chaired by the Federal Chancellery / - / All elements of open science according to the 2021 Recommendation. / The document is available for download at: <https://www.bmfwf.gv.at/forschung/forschung-oesterreich/forschungsinfrastruktur.html> (in German).

(4) Implementation framework for the EU missions of Horizon Europe in Austria (“Mehr Lebensqualität und Nachhaltigkeit durch Forschung und Anwendung. Umsetzungsrahmen für die EU-Missionen von Horizon Europe in Österreich”) / The implementation framework contributes to the tailored, effective and verifiable realisation of the five EU missions of Horizon Europe in Austria. / The Federal Ministry of Education, Science and Research and the Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology led the development of the framework. / - / Open Science is a legal obligation under Horizon Europe. The document is available for download at: https://www.bundeskanzleramt.gv.at/themen/forschungskoordination_fti.html (in German).

(5) Digital Skills Initiative for Austria (“Strategie Digitale Kompetenzen Österreich”) / The initiative aims to improve basic digital skills in the population and IT skills for the economy. While primarily focused on advancing digital literacy, the initiative also complements the broader objectives of open science by promoting skills essential to digital research and collaboration / The initiative is supported by the Federal Chancellery of Austria, the Federal Ministry for Arts, Culture, Civil Service and Sport, the Federal Ministry of Labour and Economy, and the Federal Ministry of Education, Science and Research / - / - / Link:

<https://www.digitalaustria.gv.at/dam/jcr:e84a42c3-f2e7-4642-9ca0-76d7e8c61216/Strategie-Digitale-Kompetenzen-Oesterreich-PDF-UA-1.pdf> (in German).

- 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: [\(ii\) 17.a, 17.b, 17.c, 17.f, 17.h](#))**

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organisations that are involved in the development and implementation of the plan, strategy or the roadmap.

Austria adopted its Open Science Policy in 2022. The “Open Science Policy Austria” was collaboratively developed by the Federal Ministry of Education, Science and Research, the Federal Ministry for Digital and Economic Affairs, and the Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology. With this policy, Austria demonstrates its strong commitment to the Open Science movement and the European Open Science Cloud (EOSC). The Austrian Open Science Policy is based on international open science principles as recommended by the Open Science Policy Platform, a group of experts that supports and advises the European Commission in developing strategies and implementing the European Agenda for Open Science in practice (https://research-and-innovation.ec.europa.eu/strategy/strategy-research-and-innovation/our-digital-future/open-science_en#the-eus-open-science-policy). The national policy also considers the 2021 UNESCO Recommendation on Open Science. The Austrian Open Science Policy is available at: <https://www.bmfwf.gv.at/wissenschaft/leitthemen/digitalisierung/open-science/open-science-policy-austria.html>

- 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at institutional level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: [\(ii\) 17.d, 17.e, 17.g](#))**

YES / UNDER DEVELOPMENT / NO

If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and [elements of open science](#)**Error! Bookmark not defined.** included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

In Austria, various approaches are being pursued to promote Open Science at the institutional level. Some examples include: (1) Research data management: 11 universities in Austria have policies for research data management in place, encouraging various open science practices and the adoption of the FAIR principles: Medical Universities of Graz, Innsbruck, Vienna; Graz University of Technology; TU Wien; Universities of Linz, Graz, Innsbruck, Vienna; University of Music and Performing Arts Vienna; Vienna University of Economics and Business. Policy example: <https://rdm.univie.ac.at/rdm-policy-and-faq/>. (2) Open Access: Several universities have developed open access policies. Policy example: <https://openaccess.univie.ac.at/en/oa-basics/university-of-vienna-oa-policy/>. Furthermore, the university libraries of the universities of Vienna, Graz, Innsbruck and Klagenfurt have set up open access repositories. In addition, the University of Vienna, for example, has negotiated six open access agreements at the institutional level: <https://openaccess.univie.ac.at/en/funding/oa-publishing-agreements/>. (3) Open educational resources (OER): The universities of Vienna, Graz, Innsbruck and Graz University of Technology have set up OER repositories in addition to OER policies. The University of Vienna has also launched the OERhub, a platform that bundles open educational

resources from several Austrian repositories: <https://oerhub.at/en>. OER policy example: <https://oer.univie.ac.at/oer-policy-der-universitaet-wien/>. Additionally, the University College of Teacher Education Lower Austria, has developed guidelines for creating and using OER: [Open Educational Resources Policy | Pädagogische Hochschule Niederösterreich](#). Forum New Media Austria (FNMA) has prepared "Recommendations for the Integration of Open Educational Resources at Universities in Austria": <https://fnma.at/>. The project "Open Education Austria" for the joint development of a national OER infrastructure has published a guideline for the creation of OER, which includes the definition of OER according to UNESCO's 2019 Recommendation: <https://oer-portal.uni-graz.at/edu-sharing/components/render/58f36d33-3cef-4631-a2cf-617b864efde9>. The Federal Ministry responsible for science and research also provides for the expansion of shared OER services in the higher education sector. In the current period of the performance agreements (2025-2027) with all public universities, the ministry has therefore anchored the development of an open, trustworthy digital infrastructure through the further development of and participation in national shared services for OER/MOOCs and the expansion of the range of interactive teaching materials for self-study. (4) Stakeholder platforms: The Austrian University Conference (uniko) has set up the interdisciplinary stakeholder platform Open Science Austria (OSA), which is dedicated to information and dissemination, coordination and networking as well as the development of recommendations and guidelines in the areas of Open Science: <https://www.osa-openscienceaustria.at/en/>. (5) Citizen Science: The Citizen Science Network Austria, with the associated online platform "Österreich forscht", is a network of institutions supporting the establishment and promotion of citizen science in Austria: <https://www.citizen-science.at/en/>.

2.6 In your country, are there specific funding mechanisms^{vii} for open science at the national level? (ref.: (ii) [17.i](#), (iii) [18.i](#), (iv) [19.c](#))

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

Austria has specific funding mechanisms at the national level to support open science practices. These mechanisms ensure compliance with open science principles such as open access, data sharing, and transparency in research processes: (1) Austrian Science Fund (FWF): (a) Open Access Support: the FWF mandates that publications resulting from funded projects be openly accessible. It provides funding to cover costs associated with publishing in open access journals or repositories. (b) Data Management Plans (DMPs): FWF requires DMPs for research projects and supports costs associated with implementing open data practices. (c) open science Infrastructure: FWF funds initiatives that support the development of open access infrastructures and tools, e.g. repositories such as arXiv or Europe PMC, and databases such as DOAJ or OAPEN. (d) Citizen Science: FWF funds Citizen Science projects through its dedicated funding programme. Websites: <https://www.fwf.ac.at/en/about-us/what-we-do/open-science>. (2) Austrian Research Promotion Agency (FFG): (a) Some FFG funding programmes include requirements for Open Science, particularly for collaborative projects that emphasise FAIR data principles (Findable, Accessible, Interoperable, and Reusable). (b) The FFG also supports the integration of open science practices in public-private research collaborations. Website: <https://www.ffg.at/en>. (3) European Open Science Cloud (EOSC): Austria participates actively in EOSC, which provides funding opportunities for Austrian researchers to align with European open science standards, fostering collaboration and the development of shared infrastructures. Website: <https://eosc-austria.at/>. (4) Sparkling Science 2.0: The research funding programme aligns closely with the 2021 UNESCO Recommendation on Open Science by promoting innovative, participatory strategies at institutional levels. It is funded by the Federal Ministry of Women, Science and Research and coordinated by the Agency for Education and Internationalisation (OeAD). The programme is focused on fostering Citizen Science and involves research facilities, schools as well as partners from business and society in all stages of the research cycle. Website: <https://oead.at/en/study-research->

teaching/citizen-science/sparkling-science. (5) As part of the university funding for the period 2019–2021, the Austrian Federal Ministry of Education, Science and Research dedicated EUR 50 million to the topic of “Digital and social transformation”, including the area of Open Science. As part of a call for proposals, 34 projects with a focus on enabling digital and/or social innovation(s) at public universities were selected for implementation between 2020 and 2024. Five of these were related to Open Science. The brochure “Digital and Social Transformation” (2020) provides more information on these projects. It is available for download at: https://pubshop.bmbwf.gv.at/index.php?article_id=9&sort=title&search%5Bcat%5D=83&pub=799 (in German). For instance, the AT2OA² network (Austrian Transition to Open Access 2) comprises 24 research institutions and promotes open access publishing for scientific publications: <https://www.at2oa.at/en/>.

2.7 In your country, have open science practices^{viii} been integrated in existing research funding mechanisms?

YES /NO

If yes, please provide details and links as relevant.

Open science practices have been integrated into existing research funding mechanisms in Austria. This integration is facilitated through various funding agencies and policies, with a particular emphasis on open access, data sharing, and FAIR data principles. Key examples include: (1) Austrian Science Fund (FWF): Open Access Policy: (a) All research outputs funded by FWF are required to be made publicly accessible via open access. (b) FAIR Data Principles: The FWF encourages and mandates adherence to FAIR data principles in funded projects, including the inclusion of data management plans and open data sharing. (c) Citizen science: FWF offers the “Top Citizen Science Program” to expand ongoing funded projects by including citizen science. Websites: <https://www.fwf.ac.at/en/about-us/what-we-do/open-science/open-access-policy>; <https://www.fwf.ac.at/en/funding/portfolio/communication/top-citizen-science>. (2) Austrian Research Promotion Agency (FFG): (a) Integration in Horizon Europe Projects: The FFG incorporates open science principles into its funding guidelines for projects, including the requirement for open access to publications and data as part of Horizon Europe projects. (b) Open Science Incentives: Support for projects that embrace open science practices, such as open data and public engagement in research. Website: <https://www.ffg.at/en>; (3) Participation in EOSC: Austria supports the European Open Science Cloud (EOSC), integrating open science practices into the national research funding landscape, encouraging the use of EOSC infrastructure for data sharing and collaboration: <https://eosc-austria.at/>. (4) Vienna Science and Technology Fund (WWTF): At the Federal State level, the Vienna Science and Technology Fund (WWTF), e.g., released a new Open Science Policy with focus on the provision of access to publications and to shareable research data in 2022: https://wwtf.at/upload/wwtf_open-science-policy_09032022.pdf. (5) “Sparkling Science 2.0”: The research funding programme funds citizen science projects. In line with an open access policy, the final reports are published on the Austrian Exchange Service (OeAD) project website. Developed teaching and learning materials must be digitally archived and made publicly available after the project ends (e.g. on www.eduthek.at/en). Website: <https://oead.at/en/study-research-teaching/citizen-science/sparkling-science/projects/overview>

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: [\(ii\) 17.i](#))

YES / **NO**

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Efforts to foster equitable public-private partnerships on Open Science in Austria are currently limited. Challenges include: (1) Lack of Dedicated Frameworks: There is no comprehensive framework specifically designed to facilitate equitable public-private partnerships on Open Science. (2) Alignment of Objectives: Aligning the goals of private sector stakeholders with the principles of Open Science – such as transparency, data sharing, and public accessibility – remains challenging. (3) Resource Constraints: Limited financial and human resources have hindered the development of structured initiatives in this area.

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: [\(ii\) 17.i](#), [23.c](#))

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

A working group at the EOSC support office Austria monitors the EOSC building processes in Austria. The country report on the status quo is regularly updated: <https://zenodo.org/records/13860938> (Q3/2024). In addition, some measures related to Open Science are integrated into national STI monitoring frameworks. The “Research Infrastructure Action Plan 2030”, for instance, includes measures such as supporting open innovation, research infrastructures open for collaboration, open data, the further development of open science and research – in particular EOSC –, and supporting the further development of solutions for data storage and management, in particular for research data repositories and archives as well as for data services and open access. An interim evaluation of the progress made, as well as an ex-post evaluation of the plan, will be carried out in the future. Source: <https://www.bmwf.gv.at/forschung/forschung-oesterreich/forschungsinfrastruktur.html> (in German)

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 What is the latest official data for your country’s percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.a](#))

The latest officially calculated percentage of Austria’s gross domestic product (GDP) dedicated to research and development expenditure was 3.34% in 2024. Source: <https://www.statistik.at/statistiken/forschung-innovation-digitalisierung/forschung-und-experimentelle-entwicklung-fe/forschungsquote-globalschaetzung> (in German).

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

Around 95% of the Austrian population used the internet in 2024. The broadband penetration stands at 200% for smartphone subscriptions, 64% for fixed broadband and 56% for mobile broadband (Q1 2024). The Broadband Austria 2030 strategy aims to provide Austria with symmetrical gigabit-capable access networks nationwide by 2030. Sources: <https://www.rtr.at/TKP/aktuelles/publikationen/publikationen/Datenvisualisierung/internet-monitor-q12024-daten.de.html> (in German), <https://www.statistik.at/en/statistics/research-innovation-digitalisation/digital-economy-and-society/ict-usage-in-households>, <https://www.ffg.at/en/Breitband2030>

3.3 Are there national research and education networks (NRENs)^{ix} active in the country? (ref.: [\(iii\) 18.c](#))

YES / NO

If **yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

ACOnet (Austrian Academic Computer Network) is the Austrian National Research and Education Network (NREN) for science, research, education, art and culture. It operates the Austrian scientific data network in close cooperation with site administrators in the Austrian federal states and continuously develops it in the interest of its participating organisations from the fields of science, research, education, art and culture. It provides a nationwide high-performance and non-overbooked fibre backbone network, offering connection PoPs in all Austrian provinces. All ACONet PoPs are connected redundantly, permitting highly stable and fail-safe network operations. With this backbone, ACONet participants can implement joint projects with high data volume requirements with minimal effort and at manageable costs. ACONet is operated by the University of Viennas' Computer Centre, in cooperation with other Austrian Universities, since 1992. Data exchange among participants and with participants of scientific networks in other countries is provided free of charge, limited only by the physical bandwidth of the participants' respective connection. ACONet has more than 250 participants from all over Austria, such as universities and other institutions of higher education, student dormitories, regional school networks, libraries and museums, hospitals, federal ministries and municipal authorities. Website: <https://www.aco.net/?L=1>

3.4 Are there national or institutional open science infrastructures in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

YES / UNDER DEVELOPMENT / NO

If **yes** or **under development**, please indicate which type(s) of infrastructure:

- **federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.**
- **community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society**
- **platforms for exchanges and co-creation of knowledge between scientists and society**
- **community-based monitoring and information systems to complement national, regional and global data and information systems.**

For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

Austria has several organisations that offer both generic and discipline-specific infrastructure services. Additionally, Austria is one of the few countries worldwide with a public research infrastructure database. The database provides information on infrastructures in science, research and industry to support their shared use. As of February 2025, 2,500 infrastructures – including open science infrastructures – were listed on the platform as open for collaboration: <https://forschungsinfrastruktur.bmfwf.gv.at/en>. As far as types of infrastructure are concerned, the following projects can be considered examples: (1) Federated information technology infrastructures: The Austrian Micro Data Center (AMDC) allows authorised research institutions access to microdata from Statistics Austria. Scientists from Austrian research institutions can apply for (fee-based) access. The use of anonymised data is restricted to

scientific purposes: <https://www.statistik.at/en/services/tools/services/center-for-science/austrian-micro-data-center-amdc>. The Cloud Infrastructure Platform (CLIP) offers an interdisciplinary, low-barrier-to-entry approach to scientific high-performance computing. It is jointly operated in Vienna by the Austrian Academy of Sciences, the Institute for Molecular Biotechnology, the Gregor Mendel Institute, and the Institute for Molecular Pathology: <https://www.clip.science/> By founding AUSSDA (The Austrian Social Science Data Archive), Austria has created a data infrastructure that offers a variety of research-supporting services, especially data archiving and assistance with data reuse: https://aussda.at/en/?no_cache=1. "Open Education Austria Advanced" was launched by Austrian universities for the joint development of a national infrastructure for Open Educational Resources (OER). The project links services from e-learning centres, central IT services, and the partner universities' libraries: <https://www.openeducation.at/en/>. The project "Open Data - Shared RDM" (under development) aims to create a framework to offer tools and infrastructures in the field of RDM as shared services for selected Austrian universities and research institutions. This promotes interoperability, standardisation and the connection with international initiatives: <https://forschungsdaten.at/en/sharedrdm/>. Austria has an internationally competitive large-scale infrastructure in cognitive neuroscience. The Austrian NeuroCloud (ANC) is a FAIR-enabling open repository and has created an open environment for the storage, management and analysis of neurocognitive data across locations: <https://anc.plus.ac.at/>. (2) Community managed infrastructures / platforms for exchanges and co-creation of knowledge: "Österreich forscht" presents citizen science projects from all over Austria, encouraging public participation. In addition, general information on citizen science, references to literature on citizen science and event announcements may be found on the platform, as well as the network's working groups. At the end of 2024, more than 90 active projects were listed, more than 600 blog posts had been published, and over 30 podcast episodes had been produced: <https://www.citizen-science.at/en/>. The Vienna Museum of Technology offers a new research platform to make its entire collection available online. The online database currently comprises about 158,000 object datasets and roughly 196,000 archive datasets: <https://www.technischesmuseum.at/museum/online-collection>. The Open Innovation in Science (OIS) Center of the Ludwig Boltzmann Gesellschaft (LBG) offers support and consulting services to enable scientists and scientific research institutions to involve relevant stakeholders and integrate their knowledge into research activities. The Centre focuses on open and collaborative research practices in connection with the involvement of citizens and interest groups (e.g. patients and patient representatives) in the health sector to promote the social impact of scientific research: <https://ois.lbg.ac.at/?lang=en>. (3) Community-based monitoring and information systems: The "Austrian Datahub for Open Access Negotiations and Monitoring" prepares publication data from various sources (e.g. FIS/CRIS systems) to enable their use for Austria-wide open access monitoring and to support negotiations with scientific publishers. The Datahub was developed and implemented within the framework of the project AT2OA² (2021–2024) which aims to advance the transformation from closed to open access in scientific publications: <https://oamonitor.obvsg.at/> (in German).

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: [\(iii\) 18.e, 18.f](#))

YES / NO

If **yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science¹, stage of research, accessibility to all the relevant users in the region or internationally).

The Austrian Research Infrastructure Database provides an information platform on research infrastructures in science, research and industry: <https://forschungsinfrastruktur.bmfwf.gv.at/en>. MEDem (Monitoring Electoral Democracy), for instance, is a European research infrastructure based at the University of Vienna, Austria, that

aims to harmonise and integrate democracy research data across Europe in line with open science and FAIR Data principles. MEDem supports the EOSC integration by linking its services to CESSDA and SSHOC, thereby enhancing data interoperability, accessibility and reusability. It provides automated data harmonisation, linking of key databases, and ontology-based data discovery, facilitating transdisciplinary research and open access. MEDem is supported by over 60 institutions and 850 researchers in more than 40 countries, with national coordinators from 31 European countries promoting engagement. By democratising access to high-quality democracy data, MEDem promotes transparency, evidence-based policymaking and inclusive democratic dialogue and strengthens open science in democracy research: <https://www.medem.eu/medem-endorsements/>

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimise infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: [\(iii\) 18.g](#))

YES / NO

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science), stage of research, accessibility to all the relevant users in the region or internationally).

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: [\(iv\) 19.a, 19.c](#))

YES / NO

If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

At the institutional level, there are various examples of systematic capacity building on open science, such as the following: (1) The University of Innsbruck is offering an Open Science Summer School (formal) as well as a university course “Data Librarian” (formal): <https://www.uibk.ac.at/en/weiterbildung/economics/open-science/>; <https://www.uibk.ac.at/en/weiterbildung/culture-communication/data-librarian>. (2) TU Wien participated in the development of a joint training program to strengthen open science and research data management (formal) and offers an “Introduction into Research Data Management” for bachelor and master students (lecture; formal): <https://rdmtraininghub.eu/>. (3) Capacity-building activities related to open data/research data management at the University of Vienna include establishing a university-wide RDM Network and a Data Stewardship programme (formal). (4) The University of Graz also offers a data steward certificate course: <https://forschungsdatenmanagement.uni-graz.at/de/universitaetskurs-data-steward/> (in German). In addition, the University adopted an OER policy in 2020. An institutional OER repository has been accessible and usable since 2023: https://oer-portal.uni-graz.at/start/index_en.html. The Arqus Open Science Ambassador Network is a joint expert network of open science enthusiasts constituted in 2023: <https://argus-alliance.eu/research/argus-ri/open-science/open-science-ambassador-network/>. The ambassadors promote open science, share best practices in their scientific disciplines, and give advice when needed. The network is a joint bottom-up initiative within the Arqus European University Alliance. (5) BOKU University coordinates the Citizen Science Network Austria and its associated online platform “Österreich forscht”: <https://www.citizen-science.at/en/>. Regular training on implementing citizen science in the research cycle for different target groups has been integrated into HR development courses since 2019. These courses (formal) have

recently been made available for members of other universities (e.g. TU Wien) and beyond (e.g. EURAC). (6) At the University of Veterinary Medicine Vienna, knowledge about open science and open access is integrated into information literacy education and continuous education formats for researchers and teaching staff (formal). It is also part of the university library's service portfolio. In addition, there is an ongoing informal exchange between the university's governance, the university library and research staff. (7) "wisskomm" is a Community of Practice and one of the leading voices on matters of predatory practices and quality in scholarly open-access publishing for librarians and research staff in German-speaking countries, for example, through its mailing list with about 500 subscribers and its blog, <https://in-transition.at/>. (8) In the "Research Data" digitisation cluster, several Austrian universities have joined forces with three national funding organisations – the Austrian Science Fund (FWF), the Austrian Research Promotion Agency (FFG), and the Vienna Science and Technology Fund (WWTF): <https://forschungsdaten.at/en/>. (9) Austria's Agency for Education and Internationalization (OeAD) offers various training courses, e.g. the OpenAIRE Open Science Bootcamp or webinars on various aspects of Citizen Science (formal): <https://oead.at/en/study-research-teaching/citizen-science/center-for-citizen-science/events>. (10) The Austrian National Research and Education Network AConet (cf. 3.3) provides a nationwide fibre-optic backbone with high-performance access to international academic networks via GÉANT and the Internet. Additionally, AConet offers its members various target-group-specific (formal) services to support research projects and communities. (11) The EOSC Support Office Austria (EOSC SOA) was launched as Austria's national EOSC initiative in 2021. EOSC SOA supports the coordinated development of Austrian open science policies and communities while working with local partners to integrate Austrian research data infrastructure into the broader European landscape. In addition, a collection of open science use cases, tools, templates and best practices can be found on the EOSC SOA website: <https://eosc-austria.at/>

4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: [\(iv\) 19.b](#))

YES / **NO**

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Austria is actively promoting open science through various initiatives, including the development of an Open Science Policy by the Federal Ministry of Education, Science and Research (cf. 2.2). This policy aims to foster a culture of openness and transparency in scientific research. In the realm of higher education, institutions like the University of Innsbruck are integrating open science principles into their strategic plans. The university's development plan for 2025 emphasises the importance of Open Research and the enhancement of research infrastructures, reflecting a commitment to open science practices. Additionally, Austria is advancing digital competencies through the Digital Skills Initiative, which includes the introduction of a national competence framework for digital skills: <https://www.digitalaustria.gv.at/kompetenzen/strategien.html> (in German). While this

framework primarily focuses on digital literacy, it complements the broader objectives of Open Science by promoting skills essential for digital research and collaboration. Furthermore, efforts are underway to develop an Austrian Open Educational Resources (OER) certification for higher education institutions and their employees (https://link.springer.com/chapter/10.1007/978-3-658-38703-7_9#DOI). This certification aligns with the digi.kompP model, which encompasses competencies related to creating digital materials, including aspects of open data and copyright. The development of the digi.kompP model was coordinated by the Virtual University College of Teacher Education on behalf of the Federal Ministry of Education, Science and Research in 2016 and has subsequently been revised to incorporate recent developments. While these initiatives indicate a strong movement towards incorporating open science competencies into higher education curricula, specific information regarding a standardised framework for open science competencies within research skills curricula across all Austrian higher education institutions by 2025 is not readily available. For the most accurate and up-to-date information, it would be advisable to consult official communications from the relevant ministry or individual universities' strategic plans.

- 4.4 Are open educational resources^x as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: [\(iv\) 19.d](#))**

YES / PARTLY / NO

- 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: [\(iv\) 19.e](#))**

YES / NO

If yes, please provide more information and links as relevant.

The Federal Ministry of Education, Science and Research launched the “DNAustria” initiative in 2024. It is intended to increase the visibility of science and democracy education programmes in Austria by providing a central information platform, “Entdecke.DNAustria” (“Discover DNAustria”). The platform enables the public to find various science and democracy education programmes across all Austrian federal states: <https://dnaustria.at/> (in German). In addition, Austria’s largest science communication centre will be built in Vienna by 2027. The Vienna Principles, a vision paper from 2016, developed by the OANA network (Open Access Network Austria), represents a framework for the future of scholarly communication. Principle 6, “Understandability”, for instance, calls for scholarly communication to be adapted for different stakeholder groups inside and outside of academia, making this communication more meaningful and allowing for further involvement and participation. Link: <https://viennaprinciples.org/>. The University of Innsbruck has published a position paper on open science communication: <https://www.uibk.ac.at/de/public-relations/kommunikation/open-science-communication/>. Additionally, several civil society organisations and research/higher education institutions in Austria promote a better understanding of science and the scientific process for scientific knowledge to (re-)gain significance for public discourse and decision-making. Examples: (1) The Austrian non-profit adult education institutions register over two million participants per year. More than 200,000 people attend courses and individual events offered by Austrian adult education centres in science, technology and the environment – including Open Science-related courses and events – or in politics, society and culture every year: <https://www.vhs.or.at/english>. (2) Initiated by the Austrian Academy of Sciences, the youth science communication initiative “FAEKT! – science communication for the next generation” connects scientists with young people. Activities include (a) the production of videos on quality-controlled research projects, (b) engaging content on YouTube, Instagram, and TikTok, crossing the bridge between science and entertainment, and (c) providing

supplementary materials for teachers and students aligned with the secondary school curriculum: <https://www.oeaw.ac.at/en/funding/funding-programmes/subsites/trust-science-communication>. (3) The non-profit association “Diskurs. Das Wissenschaftsnetz” was founded in 2019 by scientists based in Austria. As an interface between scientific knowledge and mass media, “Diskurs. Das Wissenschaftsnetz” strengthens Open Science by making research findings widely accessible. Building on an extensive network of scientists, it regularly organises (online) press conferences and publishes press releases in which scientists from universities and other research institutions present new evidence on a broad range of topics: <https://www.diskurs-wissenschaftsnetz.at/> (in German). (4) To increase the visibility and dissemination of scientific knowledge at the local level, the University of Graz initiated diverse off-campus projects and formats. Among these projects was a pop-up store in the city centre in 2022, where more than 150 science communication events with over 2,500 visitors were held. Other projects targeted more specific audiences, bridging their interests with science communication, e.g. “Science meets Poetry” in the House of Literature (“Literaturhaus Graz”) or “Science & Cinema” in a public movie theatre. The Center for Society, Knowledge and Communication is dedicated to science communication and societal engagement and forms an umbrella organization for various science communication offers: <https://sieben.uni-graz.at/de/> (in German)

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\)](#) [20.b](#), [20c](#), [20i](#))

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

A study commissioned by Open Science Austria (OSA) found in 2024 that open science practices to date are only marginally considered within the appointment process of new professors at Austrian higher education institutions: <https://ucrisportal.univie.ac.at/de/publications/quality-assessment-in-berufungsprozessen-an-higher-education-inst> (in German). In response, Open Science Austria founded the Research Assessment (II) working group in December 2024, which is working on evaluation criteria until autumn 2026 and will also issue national recommendations by then. Initial results will be presented at the Open Science Festival 2025 at the University of Vienna. Results (i.e. recommendations) of the working group on career paths in science and research assessment (ERA Action 3 and 4) of the Austrian University Conference from 2022–2024 are available at: https://www.uniko.ac.at/modules/download.php?key=33349_DE_O&f=2&jt=7906&cs=F43B. At the level of individual higher education institutions, for example, the University of Vienna’s “Concepts on Evaluating Research Activities and Publications” already include clear recommendations to publish open access and to use persistent identifiers like DOI and ORCID iD: <https://www.qs.univie.ac.at/en/analyses/evaluation-concepts/>. Furthermore, several Austrian higher education and research institutions have joined CoARA (<https://coara.eu/agreement/signatories/?category%5B0%5D=austria#signatories>).

5.2 Have specific clauses that recognise open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in

the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.c, 20.d](#))

YES / NO

If yes, please provide details and links as relevant.

While open science practices have not yet been widely incorporated into the career evaluation and progression systems at the institutional level (cf. 5.1), there are initiatives at the national and institutional level to move in that direction: Through the performance agreements recently negotiated with them, Austrian public universities have been encouraged to widen their assessment criteria and thereby also consider signing the CoARA agreement. The CoARA agreement explicitly states the relevance of open science practices, and some higher education and research institutions in Austria have already signed it. In addition, the Open Science Austria (OSA) working group Research Assessment (II) is currently comparing and analysing national assessment criteria to which open science practices belong. The “Final report and recommendations of the Austrian Higher Education Conference Working Group on careers in research in the context of the European Research Area”, published in 2024, also calls upon all actors within the Austrian Higher Education and Research Area to engage in and value activities concerning open science: https://www.uniko.ac.at/modules/download.php?key=33349_DE_O&f=2&jt=7906&cs=F43B

5.3 Have there been any initiatives in your country to recognise and reward open science practices or are they foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h](#))

YES / NO

If yes, for each initiative, please provide more information, including on the stakeholders^{xi} that are involved or lead the initiative, and links as relevant.

In Austria, several funding opportunities recognise and reward open science practices (cf. 2.6, 2.7), for example the “Sparkling Science” research funding programme. The programme promotes collaboration between research institutions, schools and society to generate innovative scientific outcomes (<https://oead.at/en/study-research-teaching/citizen-science/sparkling-science>). Central to its goals is the opening-up of research and innovation processes towards maximally transparent and participatory working approaches as well as free access to research results and data sets. “Sparkling Science” is funded by the Federal Ministry of Education, Science and Research and is coordinated by the Agency for Education and Internationalisation (OeAD). For “Sparkling Science 2.0”, which started in 2021 and for which there is a special directive (“Sonderrichtlinie”) until 2026, calls are planned every two years, with the next one presumably opening in 2025.

5.4 Have there been any unintended negative consequences^{xii} of open science practices reported in your country? (ref.: [\(v\) 20](#))

YES / NO

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

While Austria has been proactive in promoting open science through policies like the Open Science Policy Austria, there is limited specific documentation on unintended negative consequences within the country. However, broader discussions highlight potential challenges associated with open science practices that could be relevant to Austria: (1) Legal and Copyright Issues: Implementing open access strategies necessitates careful navigation of

legal frameworks, particularly concerning copyright laws. Authors and repository operators in Austria must consider various legal provisions when making documents publicly accessible. For instance, while many publishers permit self-archiving, they often impose conditions such as embargo periods between the publication date and open online access. This underscores the importance of understanding and adhering to copyright regulations to avoid legal complications. (2) Equity and Inclusion Concerns: Open Science aims to democratise access to research, but if not implemented thoughtfully, it may inadvertently exacerbate existing inequities. For example, the "author-pays" model of open access can be exclusionary, potentially limiting participation from researchers or institutions with fewer financial resources. This dynamic can lead to stratifications in publishing, where only well-funded researchers can afford to publish openly, thereby reinforcing existing disparities. (3) Quality and Reproducibility Challenges: The push for openness must be balanced with maintaining research quality. Without rigorous peer review and quality control mechanisms, there is a risk of disseminating unverified or low-quality research. Moreover, while open science promotes transparency, it also necessitates robust frameworks to ensure that shared data and methodologies are reproducible and reliable. Addressing these challenges is crucial to prevent a potential crisis of trust in scientific research. (4) Foreign Interference: The OSA Steering Board is currently working on the topic of foreign interference and open data at Austrian universities. (5) Predatory publishing: Predatory publishing is an issue of considerable concern for academic institutions in Austria. The project AT2OA² (2021–2024) contained a subproject on predatory publishing, the first of its kind globally. As part of the project, a community of practice, "wisskomm", was developed. Today, "wisskomm" is one of the leading voices on matters of predatory practices and quality in scholarly open access publishing for librarians and research staff in German-speaking countries: <https://in-transition.at/>

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: [\(vi\)](#) [21](#))

YES, at the national level.

YES, at the institutional level.

YES, at both national and institutional levels.

NO

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant:

- **promotion of pre-prints**
- **exploration of open peer-review practices**
- **valuing and sharing of negative scientific results and associated data**
- **participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects**
- **development of participatory strategies for identifying the needs of marginalised communities and highlighting socially relevant issues to be incorporated into research agendas.**

- **promoting open innovation practices.**
- **Others (with a box to add text)**

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Austria has taken several steps at both national and institutional levels to promote innovative, participatory methods and procedural changes to increase the openness of the entire scientific process. These initiatives align with the recommended actions from the 2021 Recommendation on Open Science.

At the national level: (1) Promotion of Pre-prints: Austria has supported the use of pre-print servers for early dissemination of research, encouraging rapid sharing of findings before formal peer review. This helps make research more accessible and transparent. For example, Austria participates in the arXiv and bioRxiv networks, where Austrian researchers are encouraged to upload their pre-prints. In December 2024, the Austrian Science Fund (FWF) and nine other leading European research funding organisations signed a Statement of Intent to jointly support and further develop the Open Research Europe (ORE) publication platform: <https://zenodo.org/records/14624287>. (2) Exploration of Open Peer-Review Practices: Some national funding bodies, such as the Austrian Science Fund (FWF), encourage open peer-review practices, including open comments and transparent reviewer identities, to enhance transparency and accountability in scientific publishing: <https://www.fwf.ac.at/en/about-us/what-we-do/open-science>. (3) Valuing and Sharing Negative Scientific Results: Austrian initiatives promote the open sharing of negative results to avoid publication bias and increase the value of all scientific findings. Platforms like the Open Science Framework (OSF) and Open Research Europe facilitate this practice. (4) Promoting Open Innovation Practices: An Open Innovation Strategy for Austria was drawn up in 2016 in a broad-based participatory process involving not only stakeholders from research, science and business but also interested members of the general public. The strategy provides an analysis of the national and international status quo on open innovation while providing a vision for 2025 and concrete measures to carry out. Measure 12 of the Open Innovation Strategy tackles how to embed open data and open access principles in research. Link to the “Open Innovation Strategy for Austria”: https://openinnovation.gv.at/wp-content/uploads/2015/08/OI_Barrierefrei_Englisch.pdf. Additionally, Austria is implementing the European Council Recommendation on the Guiding principles for knowledge valorisation with its respective Codes of practice – Code of practice on citizen-engagement (2024) and Code of practice on industry-academia co-creation for knowledge valorisation (2024). Furthermore, Open innovation is supported through funding programs that encourage collaborations between researchers, industry, and society. Austrian organisations like the Austrian Research Promotion Agency (FFG) fund projects with open innovation approaches.

At the institutional level: (1) Participatory Methods and Citizen Science: Universities such as the University of Vienna and TU Wien actively promote participatory research methods, including citizen science and crowdsourcing scientific projects, where the public contributes data or research insights. The Citizen Science Network Austria initiative supports crowdsourced data collection and public participation in various scientific projects: <https://www.citizen-science.at/en/>. The “Sparkling Science 2.0” programme encourages participatory research by integrating citizen science approaches since 2021. The initiative fosters national as well as international collaborations between research institutions, schools, and society, enabling students and citizens to actively participate in scientific processes. The funding programme has also helped to build networks and working groups in which topics such as the engagement with hard-to-reach groups or the culture of error in (citizen) science are examined more closely. Results from these collaborations and discussions are publications (e.g. “Recognising failures in citizen science projects: Lessons learned”: <https://pos.sissa.it/393/007/pdf>) as well as public events for further education. More information:

https://www.zentrumfuercitizenscience.at/fileadmin/Dokumente/zentrumfuercitizenscience.at/Publikationen/Dokumentation_H2RG_Final_17032022.pdf (German), <https://oead.at/en/study-research-teaching/citizen-science/center-for-citizen-science/funding/sparkling-science-20/project-news>. (2) Development of Participatory Strategies: Some Austrian research institutions, in collaboration with local communities, explore participatory strategies for addressing the needs of marginalised groups and incorporating socially relevant issues into research agendas. (3) Exploration of Open Peer Review and Open Innovation: Austrian higher education and research institutions also experiment with open peer review and engage with open innovation, ensuring that research is not only open but also incorporates diverse societal inputs and collaboration across disciplines. These efforts at both the national and institutional levels are designed to foster a more open, inclusive, and transparent scientific process, ensuring that research is accessible, participatory, and socially relevant.

Others: Facilitation of Open Access Publishing: Austria is participating in 20 open access agreements at the national level as part of the Austrian Academic Library Consortium (<https://www.kemoe.at/english/open-access>). In addition, the University of Vienna, for instance, has negotiated six agreements at the institutional level (<https://openaccess.univie.ac.at/en/funding/oa-publishing-agreements/>). The projects "Austrian Transition to Open Access" (2017–2020) and "Austrian Transition to Open Access 2" (2021–2025) took a holistic view of open access in Austria (<https://www.at2oa.at/en/>). Existing services were perfected, and new best practices were established.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: [\(vii\) 22.a](#))

Austria actively participates in the European Open Science Cloud (EOSC), which offers funding opportunities that support Austrian researchers in aligning their research with European open science standards. This fosters collaboration and the development of shared infrastructures. Website: <https://eosc-austria.at>. Additionally, Austria is a member of several European and international research infrastructures. The Austrian Research Infrastructure Action Plan 2030 ("Österreichischer Forschungsinfrastruktur-Aktionsplan 2030") provides more details on these memberships. The plan is available for download on the website of the Federal Ministry responsible for science and research at: <https://www.bmfwf.gv.at/forschung/forschung-oesterreich/forschungsinfrastruktur.html> (in German)

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: [\(vii\) 22.b](#), [22.e](#), [22.f](#))

YES / NO

If **yes**, please provide links as relevant and more information.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science?

YES / NO

If **yes**, please provide links as relevant and more information.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES/NO

If yes, please provide more information.

The “Vienna Principles”, developed by the OANA network (Open Access Network Austria) in 2016, outline a number of challenges for the implementation of open science in general: <https://viennaprinciples.org/>. Other factors that could affect the implementation of the 2021 recommendation are listed under section 5.4.

-
- i Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).
 - ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realise the ideals of open science.
Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.
 - iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.
 - iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.
 - v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).
 - vi These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.
 - vii A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.
 - viii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.
 - ix A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.
 - x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))
 - xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.
 - xii For example, predatory behaviours, data migration, exploitation and privatisation of research data, increased costs for scientists, and high article processing charges associated with certain business models in scientific publishing

may cause inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.